

1. Administrative & Study Identification

Full Study Title: Phase Ib Dose Finding Study Assessing Safety and Activity of [177Lu]Lu-NeoB in Combination With Radiotherapy and Temozolomide in Subjects With Newly Diagnosed Glioblastoma and as a Single Agent in Recurrent Glioblastoma **Short Title/ Acronym:** Dose Finding Study of [177Lu]Lu-NeoB in Glioblastoma **Protocol Number:** Novartis Internal ID **NCT Number:** NCT05739942 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** [177Lu]Lu-NeoB (targeted radioligand therapy) and [68Ga]Ga-NeoB (PET imaging agent) **Phase of Development:** Phase Ib **Medical Monitor:** Novartis Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the recommended dose of [177Lu]Lu-NeoB by evaluating the incidence and nature of Dose-Limiting Toxicities (DLTs) as a single agent in recurrent glioblastoma and in combination with Radiotherapy (RT) and Temozolomide (TMZ) in newly diagnosed glioblastoma. **Secondary Objectives:** To evaluate the safety, dosimetry, and pharmacokinetics of [177Lu]Lu-NeoB; to assess progression-free survival (PFS) by modified RANO criteria, and overall survival (OS); and to explore the safety of the PET imaging agent [68Ga]Ga-NeoB.

2.2 Background & Rationale Glioblastoma is the most common and aggressive type of primary brain tumor, carrying a dismal prognosis and high mortality rate. The current standard of care (surgery followed by RT and TMZ) was established 20 years ago, and there is no defined standard for recurrent cases. Gastrin-releasing peptide receptor (GRPR) is highly expressed in glioblastoma. [177Lu]Lu-NeoB is a targeted radioligand therapy with a high affinity for GRPR, designed to deliver targeted radiation directly to tumor cells while sparing healthy tissue.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm, dose-finding study in two parallel groups **Blinding:** Open-label **Randomization:** N/A (Parallel group assignment based on disease status: Newly Diagnosed vs. Recurrent)

3.2 Planned Enrollment & Duration Target \$N\$: Up to 48 participants (27 in Group 1 [Newly Diagnosed]; 21 in Group 2 [Recurrent]) **Number of Sites:** Global Multicenter **Estimated Study**

Start: 2023 **Estimated Primary Completion:** TBD (Follow-up up to 5 years)

4. Subject Selection (Eligibility Criteria)

- 4.1 Inclusion Criteria** 1. Age $\geq$ 18 years. 2. Confirmed diagnosis of newly diagnosed glioblastoma (Group 1) OR first/second glioblastoma recurrence after standard therapy (Group 2). 3. For recurrent cases (Group 2): Presence of [68Ga]Ga-NeoB uptake at the tumor region confirmed by PET/CT or PET/MRI. 4. For recurrent cases (Group 2): Evidence of recurrent disease demonstrated by modified Response Assessment in Neuro-Oncology (mRANO) criteria, with measurable contrast-enhancing lesions.
- 4.2 Exclusion Criteria** 1. Lack of target expression (negative [68Ga]Ga-NeoB PET scan) for the recurrent cohort. 2. Inadequate hematologic, renal, or hepatic function preventing safe administration of radioligand therapy. 3. Pregnancy or breastfeeding.
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5. Intervention & Treatment Plan

- 5.1 Investigational Regimen Dosage/Frequency:** * **Group 1 (Newly Diagnosed):** [177Lu]Lu-NeoB administered every 4 weeks (Q4W) for 6 administrations, in combination with concurrent RT and TMZ. * **Group 2 (Recurrent):** [177Lu]Lu-NeoB administered every 3 weeks (Q3W) for 6 administrations as a single agent. (*Note: Up to 4 additional doses may be administered if tolerated and the patient is deriving clinical benefit, for a maximum of 10 doses*) **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Active treatment phase is 24-40 weeks.
- 5.2 Concomitant & Prohibited Therapy Permitted:** Standard of care treatments including Radiotherapy and Temozolomide (TMZ) for the newly diagnosed cohort. **Prohibited:** Unapproved experimental antineoplastic therapies during the active study phase.
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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	2-4 Weeks	Screening Informed Consent, Medical History, [68Ga]Ga-NeoB PET imaging (for target confirmation)
2	Day 0	Baseline Cohort Assignment, First [177Lu]Lu-NeoB Administration
3-10	Q3W or Q4W	Treatment / Interim Weekly safety/efficacy assessments; MRI evaluations every 8 weeks
11-18	Up to 5 Years	End of Study / Follow-up Regular visits every 8 weeks for MRI assessments, progression monitoring, and survival tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence and nature of Dose-Limiting Toxicities (DLTs) to establish the recommended dose.

Measurement Method: Clinical safety evaluation during the specific DLT period (8 weeks for Group 1; 6 weeks for Group 2) after the first radioligand administration.

7.2 Secondary & Exploratory Endpoints * Safety, dosimetry, and pharmacokinetics of [177Lu]Lu-NeoB * Progression-Free Survival (PFS) evaluated using modified RANO criteria * Overall Survival (OS) * Safety and tumor uptake characterization of the [68Ga]Ga-NeoB PET imaging agent

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous weekly safety and efficacy assessments during the treatment phase; continuous monitoring for radiotoxicity (especially renal). **Serious Adverse Events (SAEs):** Standard 24-hour reporting timelines apply. **Data Monitoring Committee (DMC):** Utilized to review dose-escalation safety data guided by the Bayesian Optimal Interval approach.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety set for DLT calculations; Intent-to-Treat (ITT) for PFS and OS evaluation. **Sample Size Justification:** Up to 48 patients (27 newly diagnosed, 21 recurrent) are statistically justified to fulfill Bayesian Optimal Interval model requirements using consecutive cohorts of 3-6 evaluable patients. **Handling of Missing Data:** Standard survival analysis methods including censoring for progression-free and overall survival at the last known contact.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: High-frequency onsite safety monitoring during the active treatment phase; remote imaging data collection. **Audit Trail:** Validated eCRF tracking of administered radioactive doses, concomitant TMZ/RT regimens, and centralized radiological (PET and MRI) reviews.

11. Study Identifiers & Governance

Primary ID: Novartis Internal ID **Registry Number:** NCT05739942

Sponsor/Lead Organization: Novartis Pharmaceuticals **Study Title:**

Dose Finding Study of [177Lu]Lu-NeoB in Newly Diagnosed Glioblastoma and in Recurrent Glioblastoma **Official Regulatory Title:** Phase Ib Dose Finding Study Assessing Safety and Activity of [177Lu]Lu-NeoB in Combination With Radiotherapy and Temozolomide in Subjects With Newly Diagnosed Glioblastoma and as a Single Agent in Recurrent Glioblastoma

12. Clinical Framework (Glioblastoma Specific)

Primary Condition: Glioblastoma (Newly Diagnosed and Recurrent)
Diagnosis Threshold: Histologically confirmed glioblastoma with target confirmation via [68Ga]Ga-NeoB PET scan **Medical Methodology:** Targeted Radioligand Therapy (TRT) targeting Gastrin-Releasing Peptide Receptor (GRPR) **Optimization Target:** Identifying the maximum tolerated / recommended therapeutic dose

13. Study Procedures & Methodology

Management Pathway: Targeted radiation combination (TMZ+RT) for new diagnoses; Monotherapy for recurrent cases **Education Protocol (HCP):** Safe handling and administration of radiopharmaceuticals ([177Lu] / [68Ga]) **Education Protocol (Patient):** Radiation safety protocols post-administration and imaging requirements **Quality Audit Mechanism:** Dosimetry checks and Bayesian cohort dose-escalation reviews

14. Observation & Follow-Up Schedule

Total Duration: 24-40 weeks of active treatment followed by up to 5 years of survival follow-up **Onsite Visit Cadence:** Every 3 weeks (Recurrent) or 4 weeks (Newly Diagnosed) for dosing; MRI every 8 weeks **Remote Monitoring Frequency:** Intermittent safety calls between active treatment cycles **Checklist Review Interval:** Weekly toxicity and symptom assessments during the DLT observation windows

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				